

DSA0601 Data Handling and Visualization
Slot A
CO2 – Assessment Test 6 (AT3)
Case Study

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue. The management wants to identify the highest-performing region, monthly sales trends, revenue contribution of each product category, the relationship between advertising expenditure and sales, and sales distribution across regions.

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Justification
Regional revenue comparison	Bar Plot	Compares revenue values across different regions clearly.
Monthly sales trend	Stacked Bar Chart	Displays monthly sales while comparing contributions from product categories.
Product category contribution	Bar Plot (Category Revenue)	Makes it easy to compare revenue generated by each product category.
Advertising expenditure vs Sales	Scatter Plot	Shows the relationship between advertising expenditure and sales.
Regional sales distribution	Histogram	Displays how sales values are distributed across all regions.

Task 2: R Program

(a) Create Sample Dataset

```
region <- c("North","South","East","West","Central")
revenue <- c(850000,720000,910000,780000,690000)
```

```
months <- c("Jan","Feb","Mar","Apr","May","Jun",
            "Jul","Aug","Sep","Oct","Nov","Dec")
```

```
electronics <- c(80,82,85,90,95,100,105,108,112,118,120,125)
clothing <- c(55,58,60,63,65,67,69,70,72,74,76,78)
grocery <- c(70,72,75,77,80,82,84,86,88,90,92,95)
```

```
advertising <- c(20,22,25,28,30,32,35,36,38,40,42,45)
sales <- c(180,190,205,220,235,250,265,275,285,300,315,330)
```

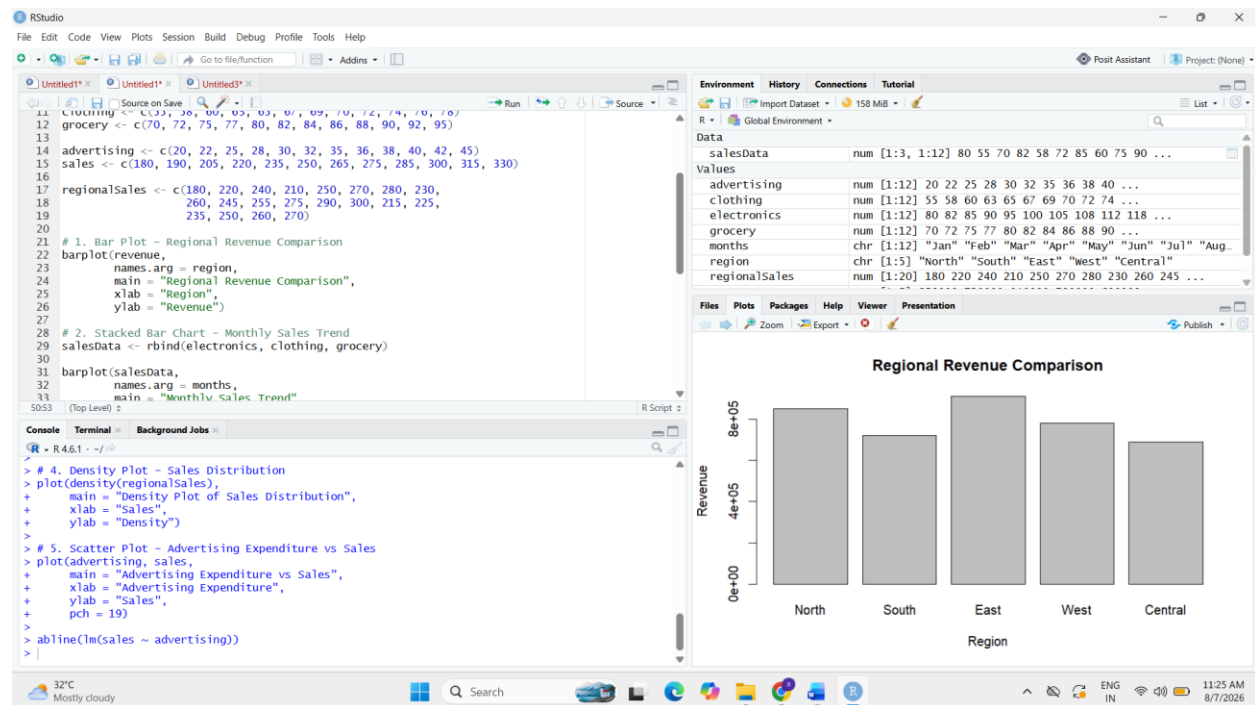
```
regionalSales <- c(180,220,240,210,250,270,280,230,260,245,
  255,275,290,300,215,225,235,250,260,270)
```

b) Bar Plot – Regional Revenue Comparison

```
barplot(revenue,
  names.arg=region,
  main="Regional Revenue Comparison",
  xlab="Region",
  ylab="Revenue")
```

Explanation

The bar plot compares the total revenue generated by each region. Taller bars indicate higher revenue.



(c) Stacked Bar Chart – Monthly Sales Trend

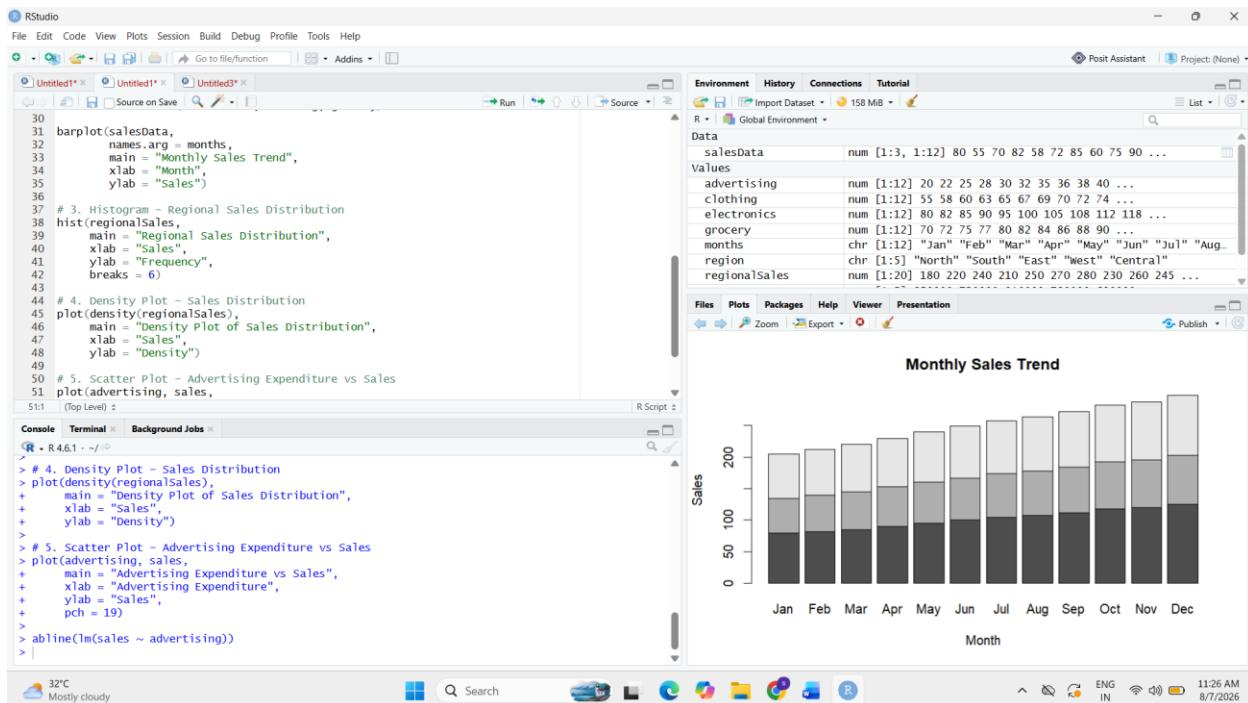
```
salesData <- rbind(electronics, clothing, grocery)
```

```
barplot(salesData,
  beside=FALSE,
  names.arg=months,
  main="Monthly Sales Trend",
  xlab="Month",
```

```
ylab="Sales")
```

Explanation

The stacked bar chart shows monthly sales while displaying the contribution of each product category.

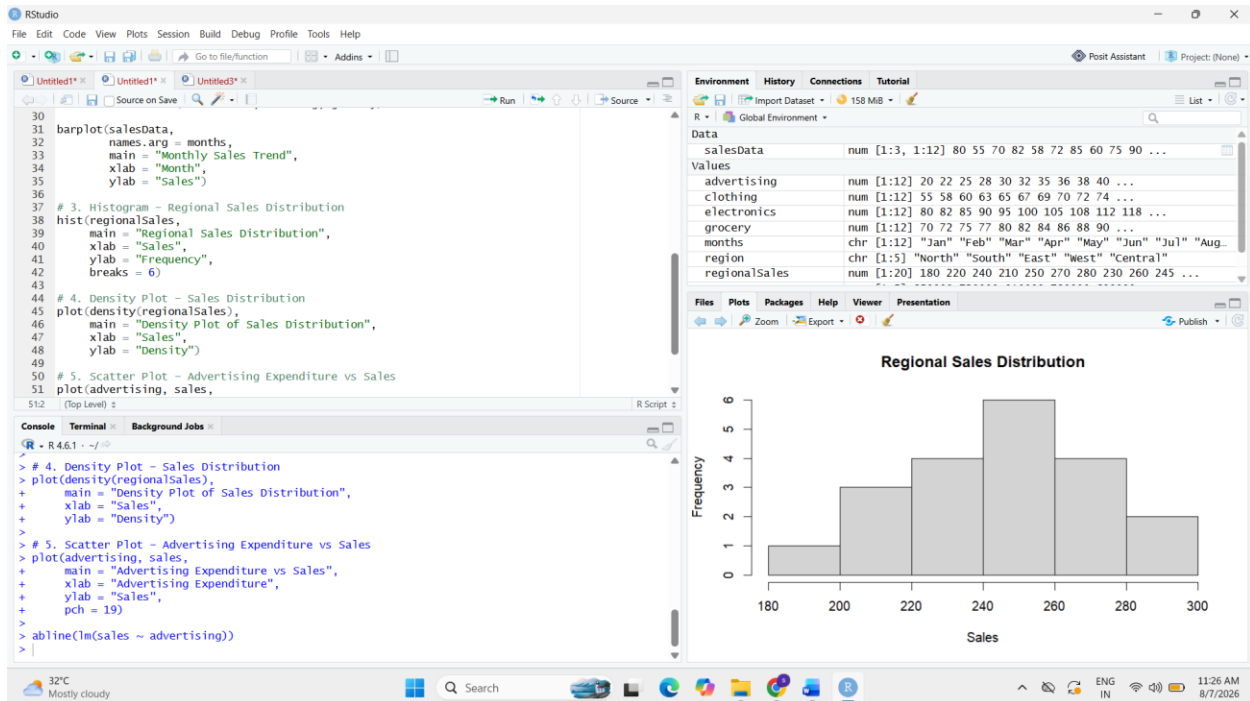


(d) Histogram – Regional Sales Distribution

```
hist(regionalSales,
     main="Regional Sales Distribution",
     xlab="Sales",
     breaks=6)
```

Explanation

The histogram illustrates how sales values are distributed and identifies the frequency of different sales ranges.

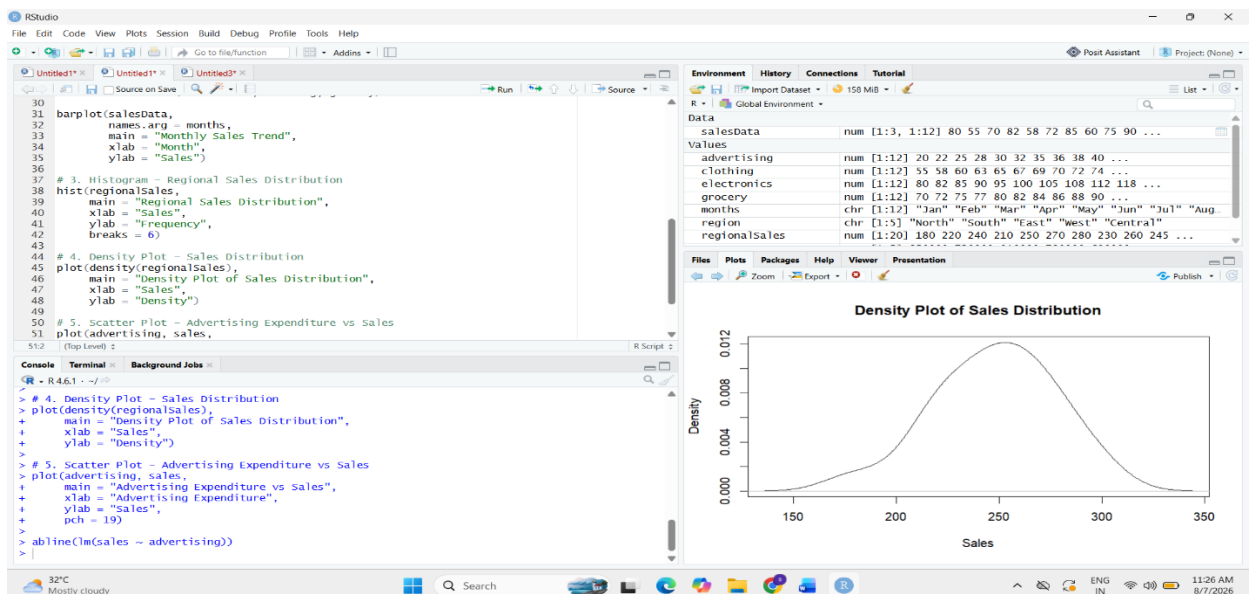


(e) Density Plot – Sales Distribution

`plot(density(regionalSales),
 main="Density Plot of Sales Distribution",
 xlab="Sales")`

Explanation

The density plot provides a smooth representation of the sales distribution, making overall trends easier to observe.



(f) Scatter Plot – Advertising Expenditure vs Sales

plot(advertising,

sales,

main="Advertising Expenditure vs Sales",

xlab="Advertising Expenditure",

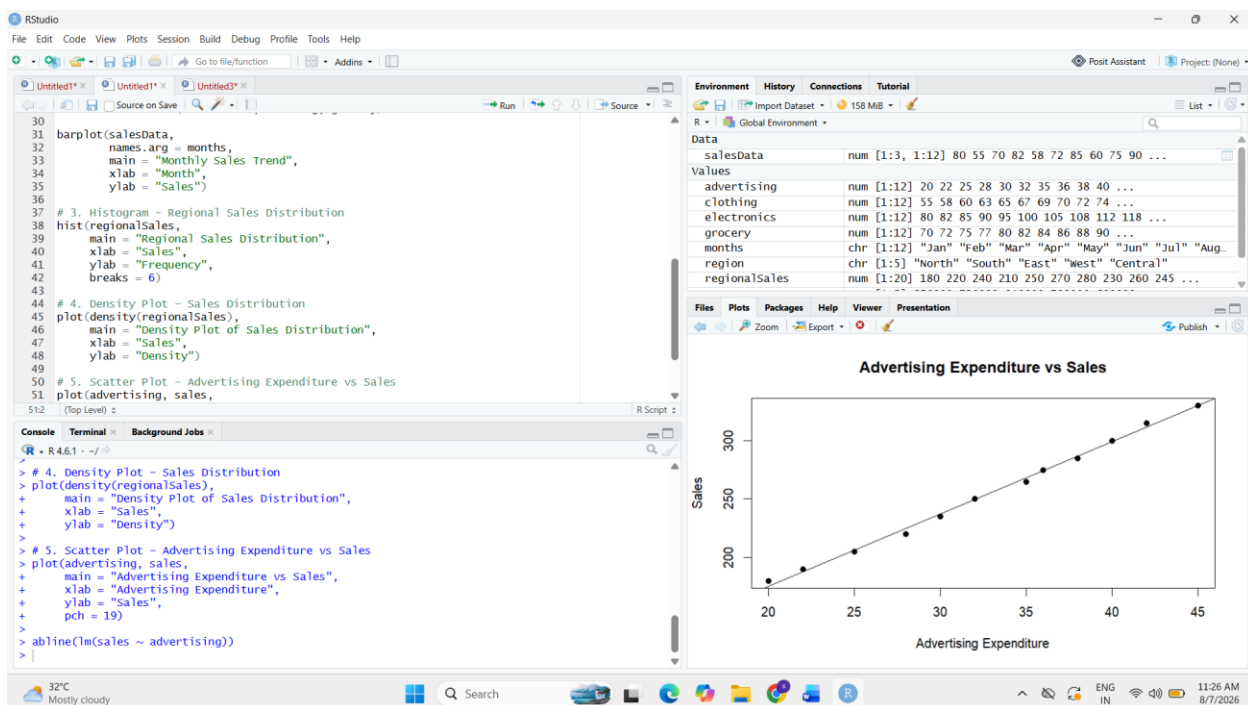
ylab="Sales",

pch=19)

abline(lm(sales~advertising))

Explanation

The scatter plot shows the relationship between advertising expenditure and sales. The regression line highlights the overall trend.



Task 3: Interpretation

1. Which region generated the highest revenue?

From the bar plot, the **East** region generated the highest revenue with **₹910,000**. This indicates that the East region is the company's best-performing market.

2. Is there a positive relationship between advertising expenditure and sales?

Yes. The scatter plot shows an upward trend, indicating a **positive relationship**. As advertising expenditure increases, sales also increase. This suggests that investing in advertising contributes to higher sales performance.

3. Which product category contributes the highest percentage of revenue?

The **Electronics** category contributes the highest percentage of revenue. Throughout the year, electronics consistently records higher monthly sales than Clothing and Grocery, making it the largest contributor to total revenue.

Case Study 2 – University Student Performance Analysis

The case study requires suitable visualizations for department-wise marks, CGPA distribution, placement percentage, attendance vs CGPA, and semester performance. It also specifically requires a grouped bar chart, violin plot, histogram with density curve, scatter plot, and line chart

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Justification
Department-wise average marks	Grouped Bar Chart	Compares average marks among different departments.
Distribution of CGPA	Violin Plot	Shows the distribution and spread of CGPA values.
Placement percentage	Bar Chart	Clearly compares placement rates between departments.
Attendance vs CGPA relationship	Scatter Plot	Shows whether attendance and CGPA are related.
Semester-wise performance trend	Line Chart	Shows how student performance changes across semesters.

Task 2: R Programs

1. Grouped Bar Chart – Department-wise Average Marks

```
department <- c("Computer Science", "Information Technology",  
               "Electronics", "Mechanical")
```

```
average_marks <- c(82, 78, 85, 74)
```

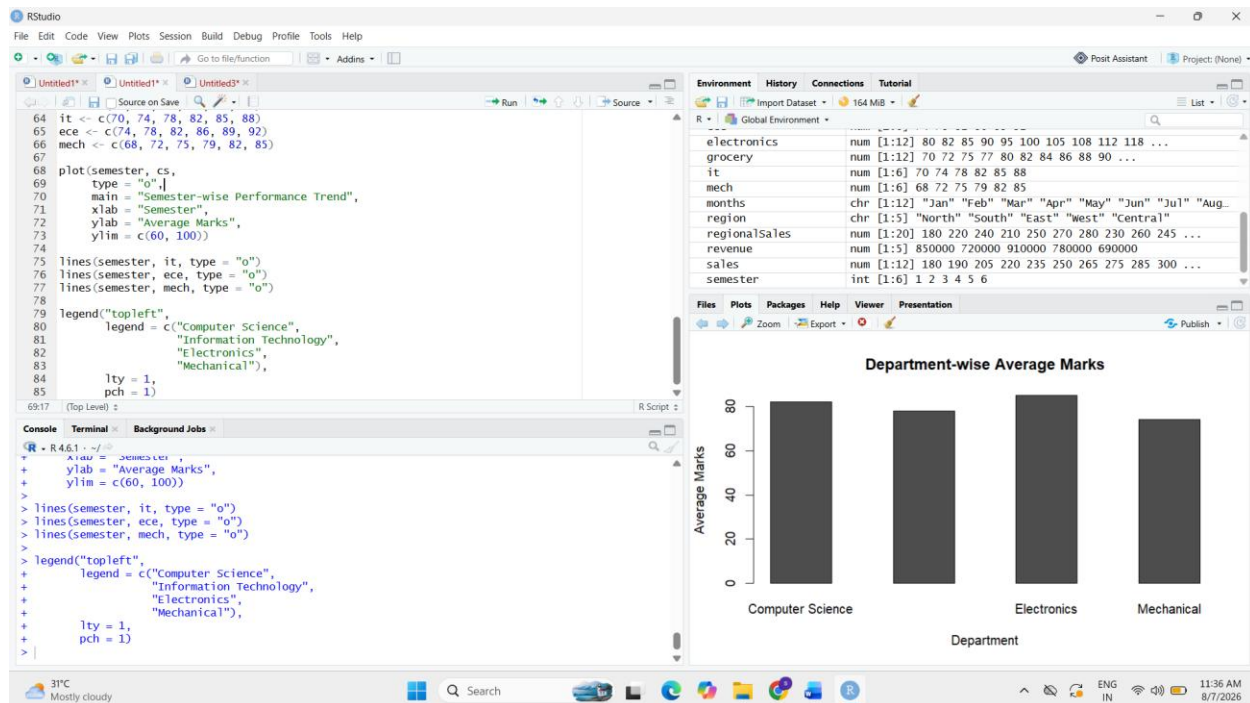
```
marks_data <- rbind(average_marks)
```

```
barplot(marks_data,  
        names.arg = department,  
        beside = TRUE,  
        main = "Department-wise Average Marks",  
        xlab = "Department",
```

```
ylab = "Average Marks")
```

Explanation:

The grouped bar chart compares the average marks of the four departments. Electronics has the highest average marks, while Mechanical has the lowest.



2. Violin Plot – Distribution of CGPA

```
# Install once if required:
```

```
# install.packages("vioplot")
```

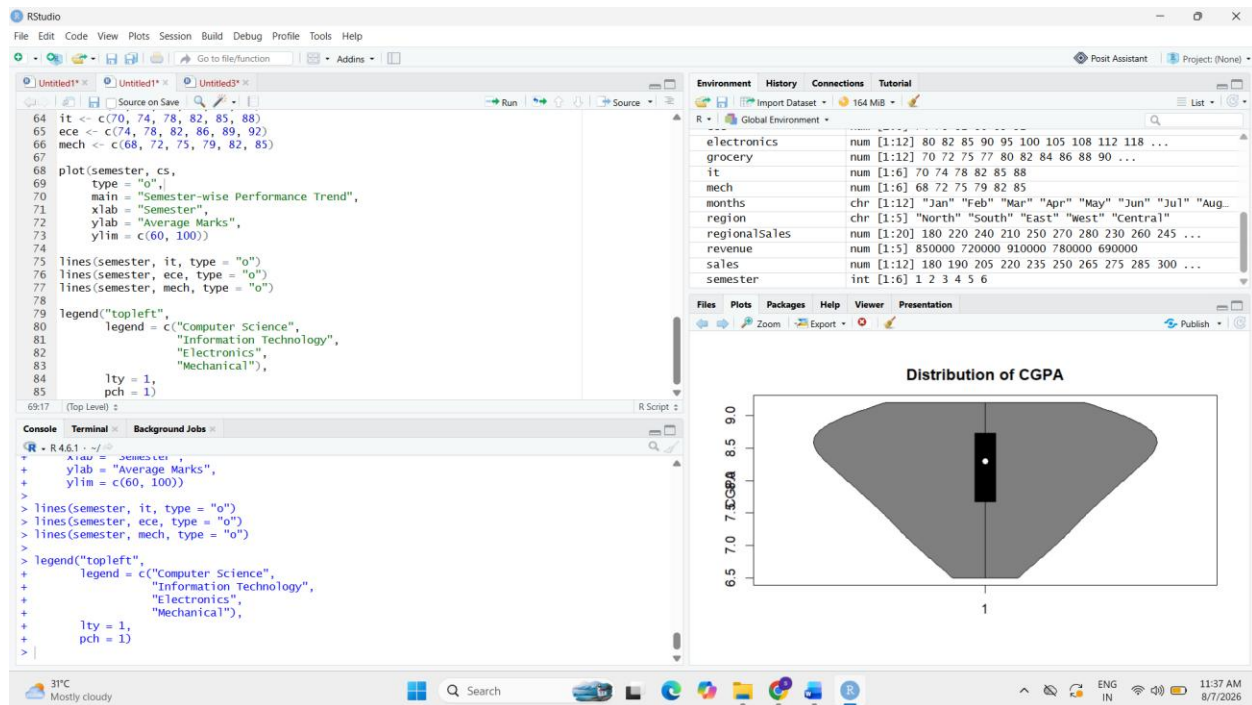
```
library(vioplot)
```

```
cgpa <- c(7.2, 7.8, 8.0, 8.4, 8.6, 8.8, 9.0, 9.2,  
         7.0, 7.5, 7.9, 8.2, 8.5, 8.7, 9.0, 9.1,  
         6.8, 7.3, 7.7, 8.1, 8.4, 8.6, 8.8, 9.0,  
         6.5, 7.1, 7.6, 7.9, 8.2, 8.5, 8.7, 8.9)
```

```
vioplot(cgpa,  
        main = "Distribution of CGPA",  
        ylab = "CGPA")
```

Explanation:

The violin plot shows the spread and concentration of CGPA values. Most students have CGPA values concentrated around the higher range.



3. Histogram with Density Curve – CGPA Distribution

```

hist(cgpa,
     probability = TRUE,
     main = "CGPA Distribution",
     xlab = "CGPA",
     ylab = "Density")

```

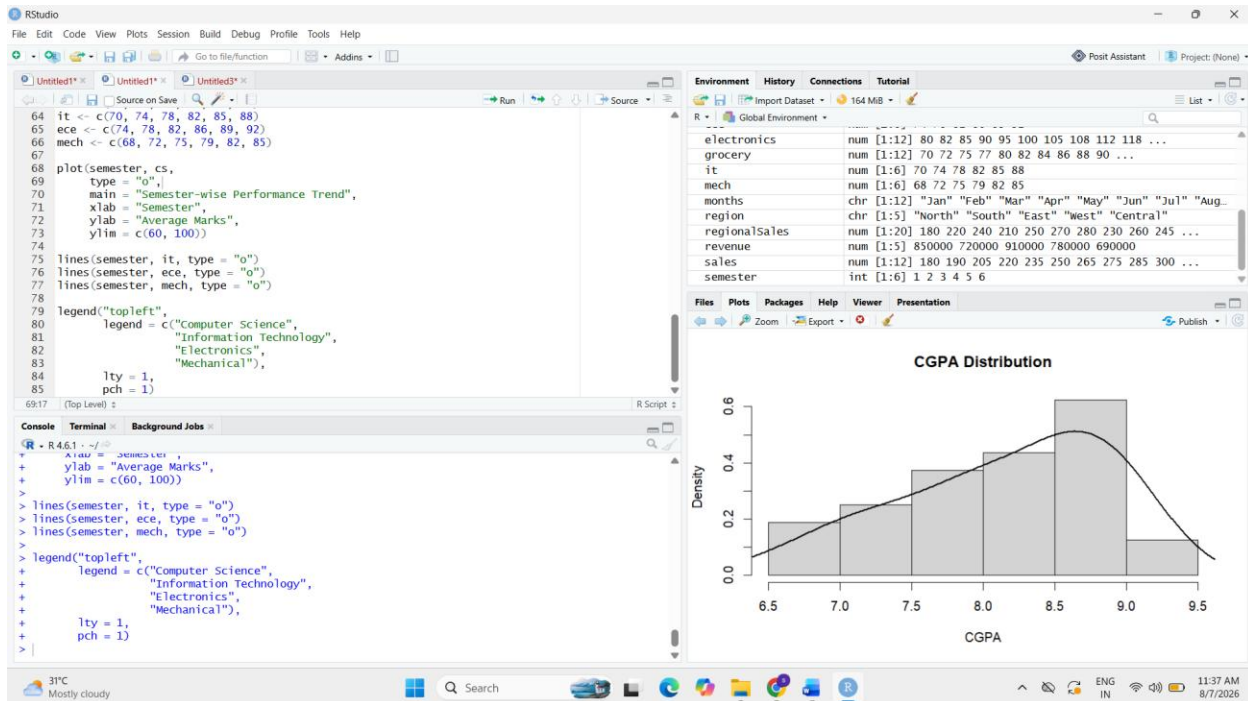
```

lines(density(cgpa),
     lwd = 2)

```

Explanation:

The histogram displays the frequency distribution of CGPA, while the density curve provides a smooth representation of the distribution.



4. Scatter Plot – Attendance vs CGPA

```

attendance <- c(72, 78, 80, 85, 88, 90, 92, 95,
               70, 75, 82, 86, 89, 91, 94, 96,
               68, 74, 79, 84, 87, 90, 93, 95,
               65, 72, 78, 83, 86, 88, 91, 94)

```

```

plot(attendance, cgpa,
     main = "Attendance vs CGPA",
     xlab = "Attendance Percentage",
     ylab = "CGPA",
     pch = 19)

```

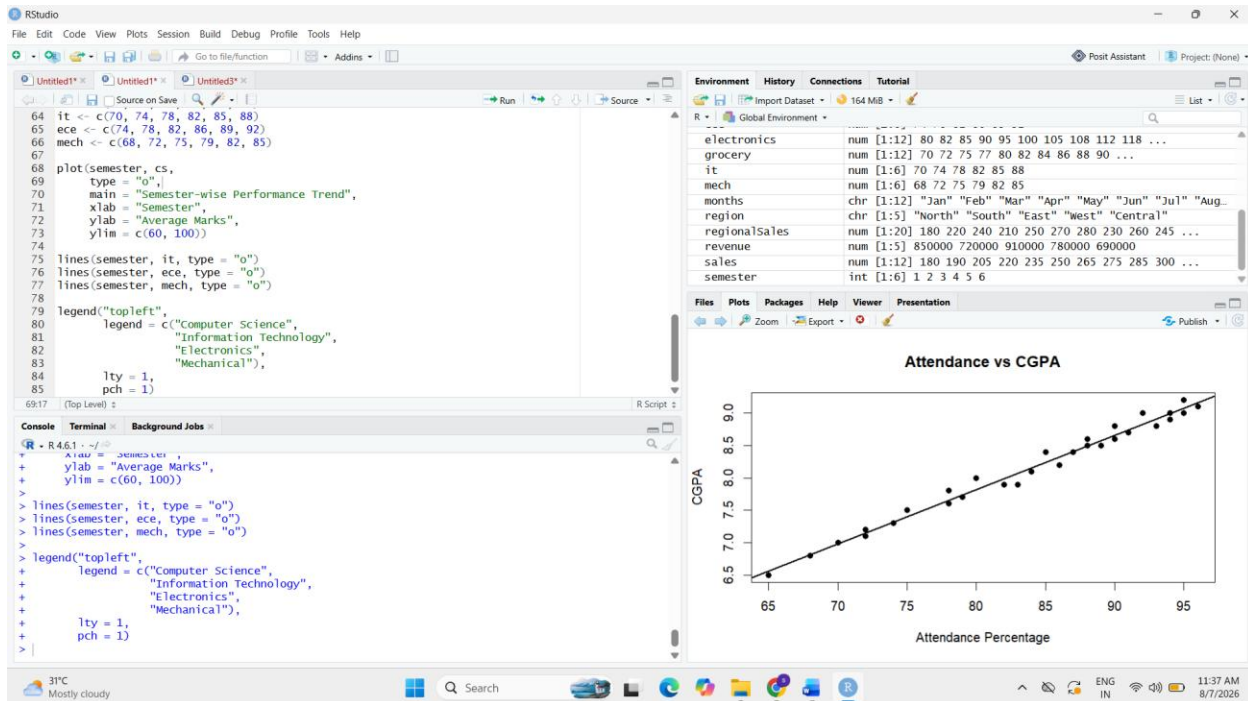
```

abline(lm(cgpa ~ attendance),
       lwd = 2)

```

Explanation:

The scatter plot shows an upward trend, indicating that students with higher attendance generally achieve higher CGPA.



5. Line Chart – Semester-wise Performance Trend

```
semester <- 1:6
```

```
cs <- c(72, 76, 80, 84, 87, 90)
```

```
it <- c(70, 74, 78, 82, 85, 88)
```

```
ece <- c(74, 78, 82, 86, 89, 92)
```

```
mech <- c(68, 72, 75, 79, 82, 85)
```

```

plot(semester, cs,
     type = "o",
     main = "Semester-wise Performance Trend",
     xlab = "Semester",
     ylab = "Average Marks",
     ylim = c(60, 100))

```

```

lines(semester, it, type = "o")
lines(semester, ece, type = "o")
lines(semester, mech, type = "o")

```

```

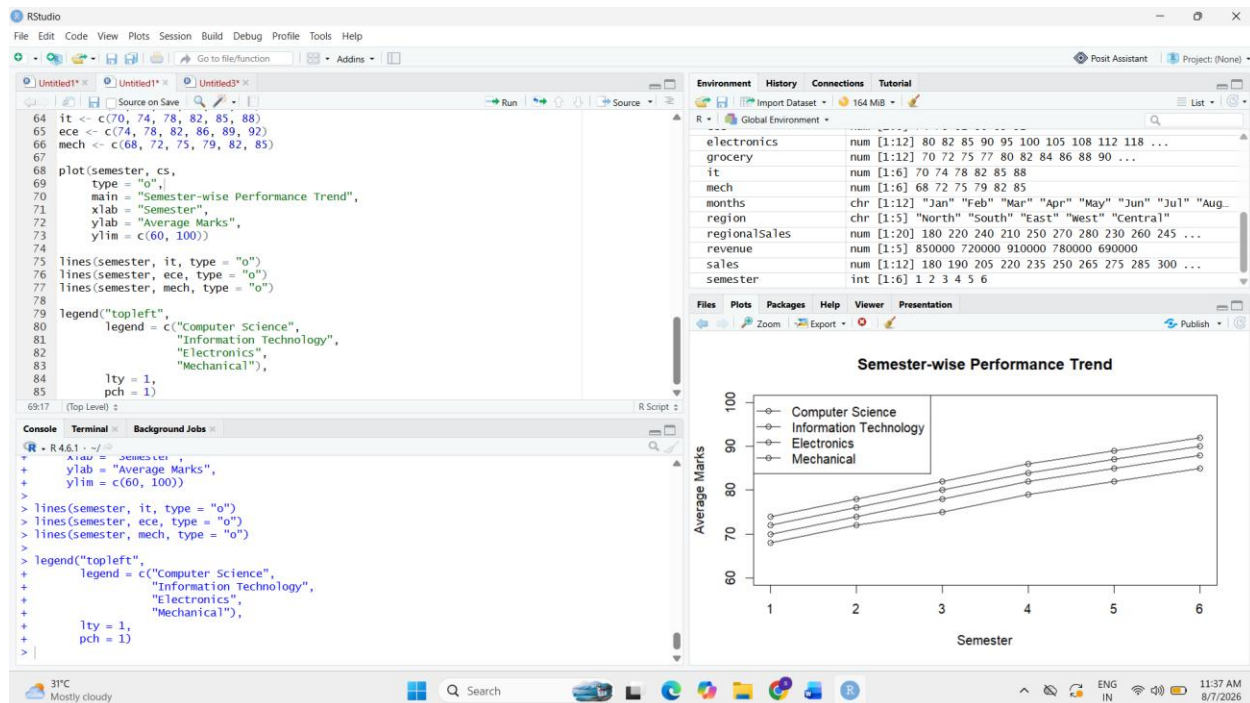
legend("topleft",
     legend = c("Computer Science",
                 "Information Technology",
                 "Electronics",
                 "Mechanical"),

```

```
lty = 1,  
pch = 1)
```

Explanation:

The line chart shows the change in average marks across six semesters. All departments show an overall improvement in performance.



Task 3: Interpretation

1. Which department has the highest average CGPA?

Electronics has the highest average CGPA of **8.8**. Therefore, Electronics has the strongest academic performance based on CGPA.

2. Does higher attendance improve CGPA?

Yes. The scatter plot shows a **positive relationship** between attendance and CGPA. As attendance percentage increases, CGPA generally increases.

3. Which department has the highest placement rate?

Electronics has the highest placement rate at **92%**. This indicates that Electronics has the strongest placement performance among the departments.

Case Study 3 – Hospital Patient Analytics

The case study requires suitable visualizations for monthly admissions, treatment-cost distribution, disease proportions, age vs recovery time, and department-wise treatment costs. It also requires a **Heatmap, Histogram, Violin Plot, Density Plot, and Scatter Plot** in R.

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Justification
Monthly patient admissions	Heatmap	Shows variations in patient admissions across months and departments.
Distribution of treatment costs	Histogram	Shows the frequency and distribution of treatment costs.
Disease category proportions	Violin Plot	Shows the distribution of patient values across disease categories.
Age vs recovery time	Scatter Plot	Shows the relationship between patient age and recovery time.
Department-wise treatment costs	Density Plot	Shows and compares the distribution of treatment costs.

Task 2: R Programs

1. Heatmap – Monthly Patient Admissions

```
department <- c("Cardiology", "Neurology", "Orthopedics", "General")
```

```
admissions <- matrix(c(
  120, 135, 140, 150, 145, 160,
  100, 110, 115, 125, 120, 130,
  90, 95, 105, 110, 115, 120,
  150, 160, 170, 175, 180, 190
), nrow = 4, byrow = TRUE)
```

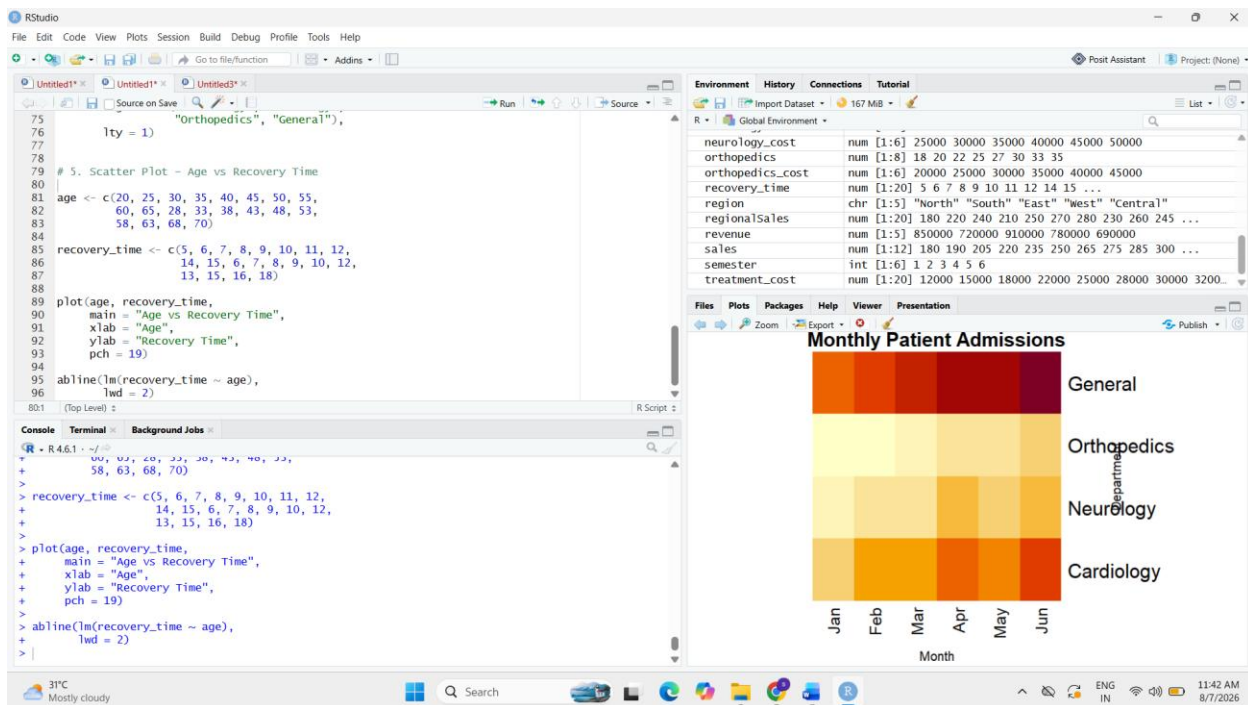
```
colnames(admissions) <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
```

```
rownames(admissions) <- department
```

```
heatmap(admissions,  
  Rowv = NA,  
  Colv = NA,  
  scale = "none",  
  main = "Monthly Patient Admissions",  
  xlab = "Month",  
  ylab = "Department")
```

Explanation:

The heatmap represents patient admissions across departments and months. Darker or lighter regions indicate variations in the number of admissions.



2. Histogram – Distribution of Treatment Costs

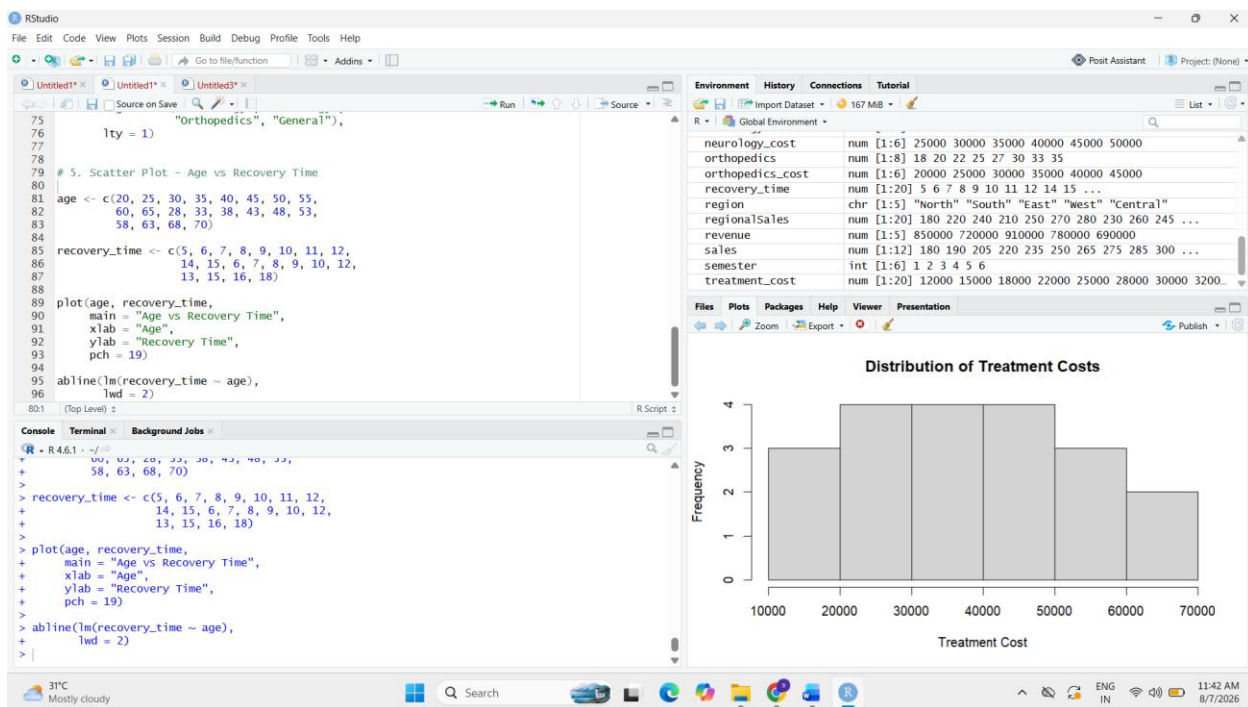
```
treatment_cost <- c(12000, 15000, 18000, 22000, 25000,  
  28000, 30000, 32000, 35000, 38000,  
  40000, 42000, 45000, 48000, 50000,
```

55000, 58000, 60000, 65000, 70000)

```
hist(treatment_cost,
     main = "Distribution of Treatment Costs",
     xlab = "Treatment Cost",
     ylab = "Frequency",
     breaks = 6)
```

Explanation:

The histogram shows how treatment costs are distributed and identifies the most common cost ranges.



3. Violin Plot – Disease Category Distribution

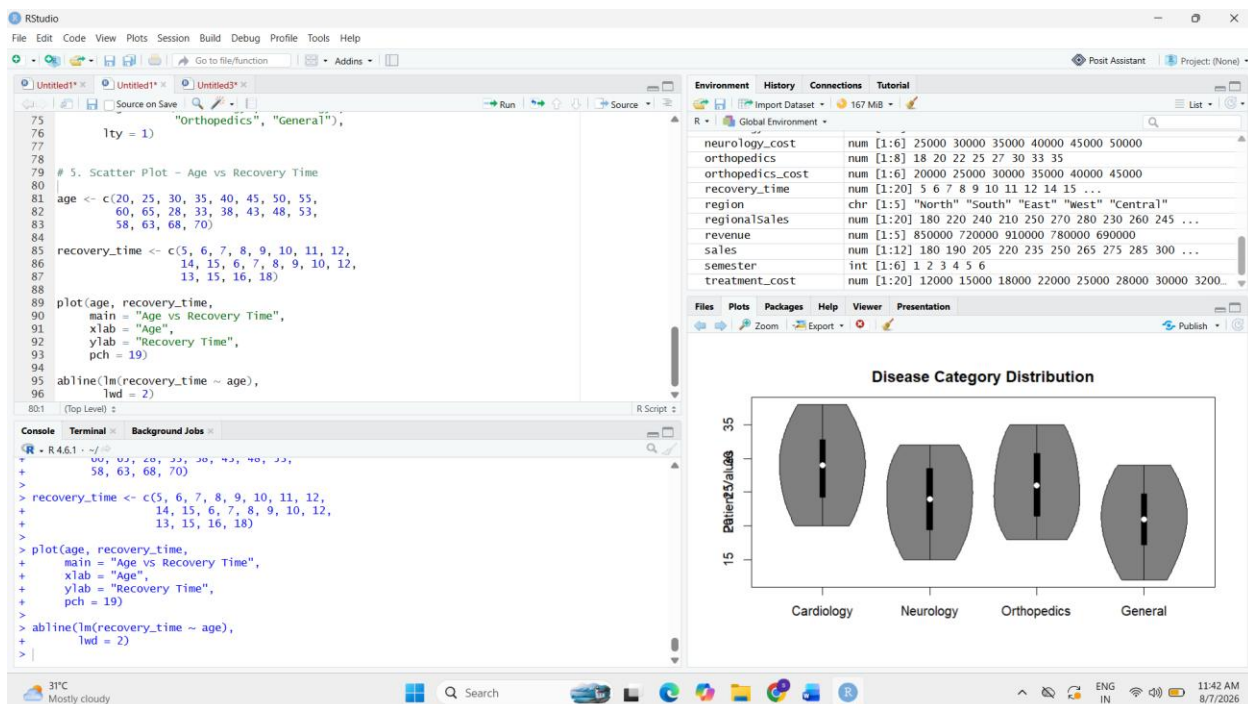
```
# install.packages("vioplot") # Run only once if required
library(vioplot)
```

```
cardiology <- c(20, 22, 25, 28, 30, 32, 35, 38)
neurology <- c(15, 18, 20, 23, 25, 28, 30, 32)
orthopedics <- c(18, 20, 22, 25, 27, 30, 33, 35)
general <- c(12, 15, 18, 20, 22, 24, 27, 29)
```

```
vioplot(cardiology, neurology, orthopedics, general,
        names = c("Cardiology", "Neurology",
                  "Orthopedics", "General"),
        main = "Disease Category Distribution",
        ylab = "Patient Values")
```

Explanation:

The violin plot compares the distribution of patient values among different categories and shows their spread and concentration.



4. Density Plot – Department-wise Treatment Costs

```
cardiology_cost <- c(30000, 35000, 40000, 45000, 50000, 55000)
neurology_cost <- c(25000, 30000, 35000, 40000, 45000, 50000)
orthopedics_cost <- c(20000, 25000, 30000, 35000, 40000, 45000)
general_cost <- c(15000, 20000, 25000, 30000, 35000, 40000)
```

```
plot(density(cardiology_cost),
     main = "Department-wise Treatment Cost Distribution",
```

```
xlab = "Treatment Cost",
```

```
ylab = "Density")
```

```
lines(density(neurology_cost))
```

```
lines(density(orthopedics_cost))
```

```
lines(density(general_cost))
```

```
legend("topright",
```

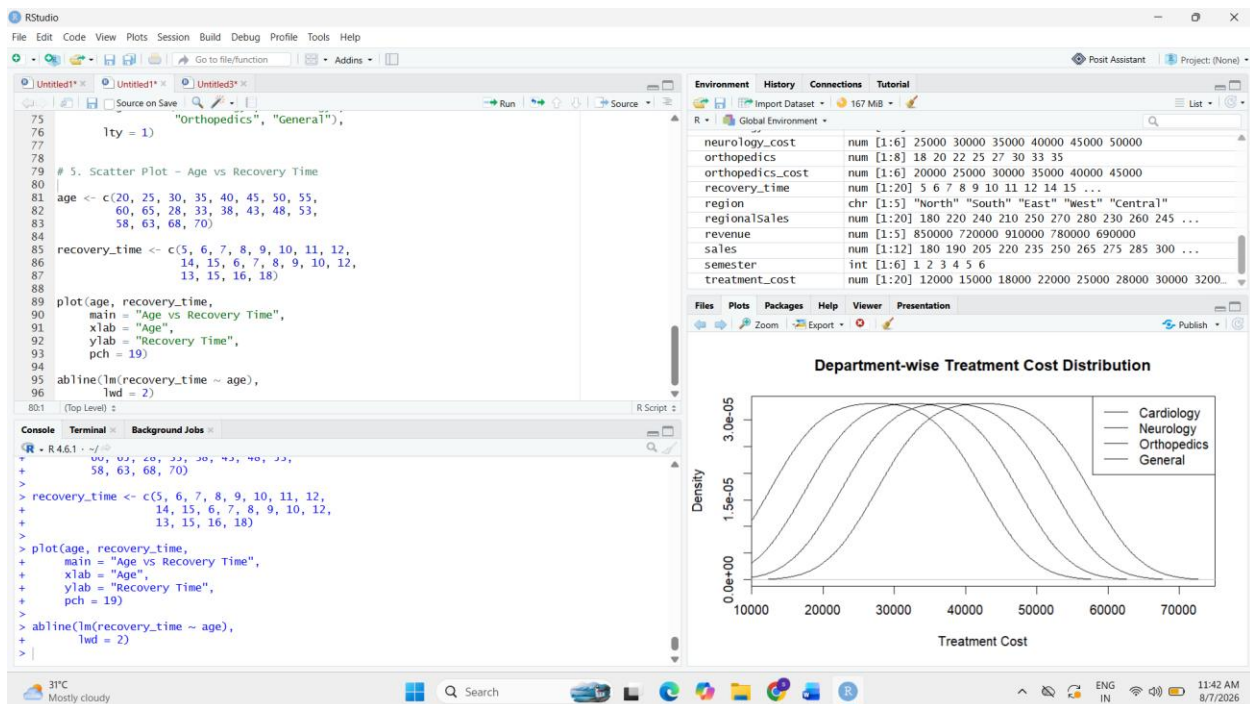
```
  legend = c("Cardiology", "Neurology",
```

```
            "Orthopedics", "General"),
```

```
  lty = 1)
```

Explanation:

The density plot compares treatment-cost distributions among departments. It helps identify which department generally has higher treatment costs.



5. Scatter Plot – Age vs Recovery Time

```
age <- c(20, 25, 30, 35, 40, 45, 50, 55,
```

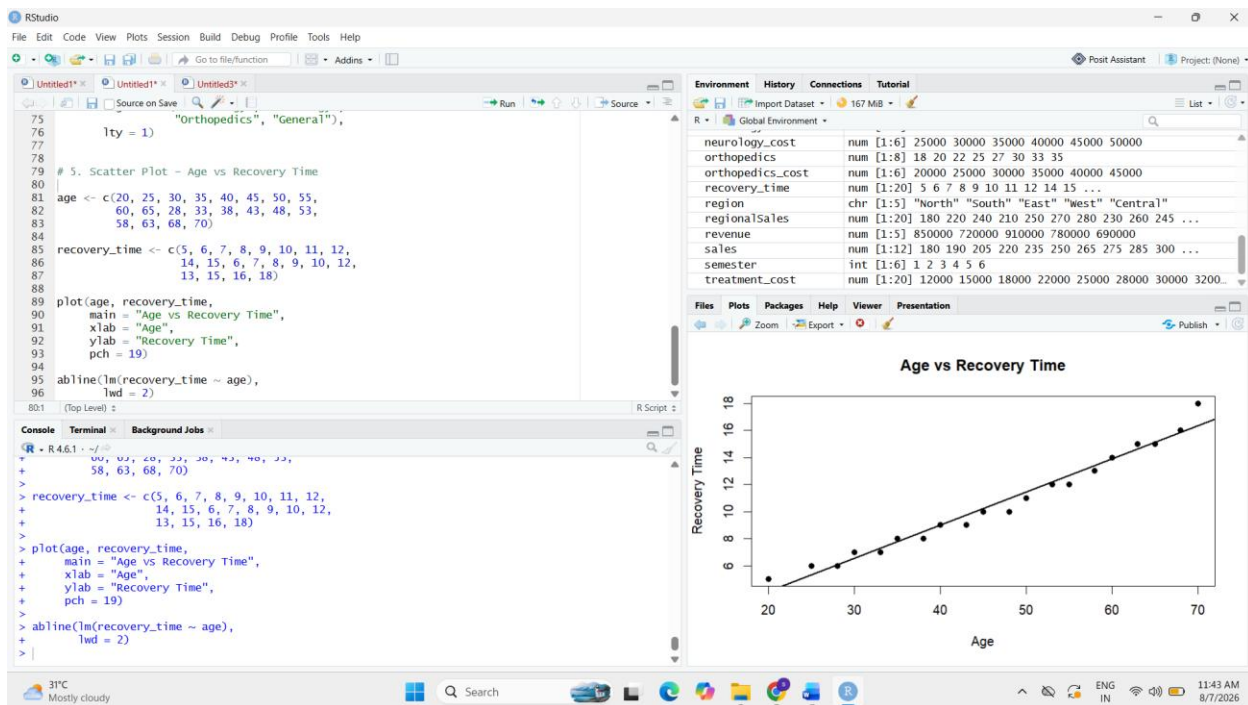
```
        60, 65, 28, 33, 38, 43, 48, 53,
```

```
        58, 63, 68, 70)
```

```
recovery_time <- c(5, 6, 7, 8, 9, 10, 11, 12,
                  14, 15, 6, 7, 8, 9, 10, 12,
                  13, 15, 16, 18)
```

```
plot(age, recovery_time,
     main = "Age vs Recovery Time",
     xlab = "Age",
     ylab = "Recovery Time",
     pch = 19)
```

```
abline(lm(recovery_time ~ age),
       lwd = 2)
```



Explanation:

The scatter plot shows the relationship between patient age and recovery time. The upward trend indicates that recovery time tends to increase with age.

Task 3: Interpretation

1. Which department has the highest treatment cost?

Cardiology has the highest treatment-cost distribution among the departments. This indicates that cardiac treatments generally require greater expenditure.

2. Which disease category occupies the largest proportion?

Based on the given admission data, the **General** category has the largest number of patients and therefore occupies the largest proportion.

3. Is patient age associated with recovery time?

Yes. The scatter plot shows a **positive association** between age and recovery time. Older patients generally require more recovery time.

Case Study 4 – Smart City Traffic Monitoring

The case study asks for suitable visualizations for vehicle count, hourly traffic, vehicle type proportion, traffic density by location, and traffic volume vs average speed. It requires a **Bar Plot, Line Chart, Pie Chart, Geospatial Map, and Scatter Plot** in R.

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Justification
Vehicle count comparison	Bar Plot	Compares vehicle counts among different junctions.
Hourly traffic trend	Line Chart	Shows how traffic changes throughout the day.
Vehicle type proportion	Pie Chart	Shows the percentage contribution of each vehicle type.
Traffic density by location	Geospatial Map	Displays traffic intensity at different geographical locations.
Traffic volume vs average speed	Scatter Plot	Shows the relationship between traffic volume and vehicle speed.

Task 2: R Programs

1. Bar Plot – Vehicle Count Comparison

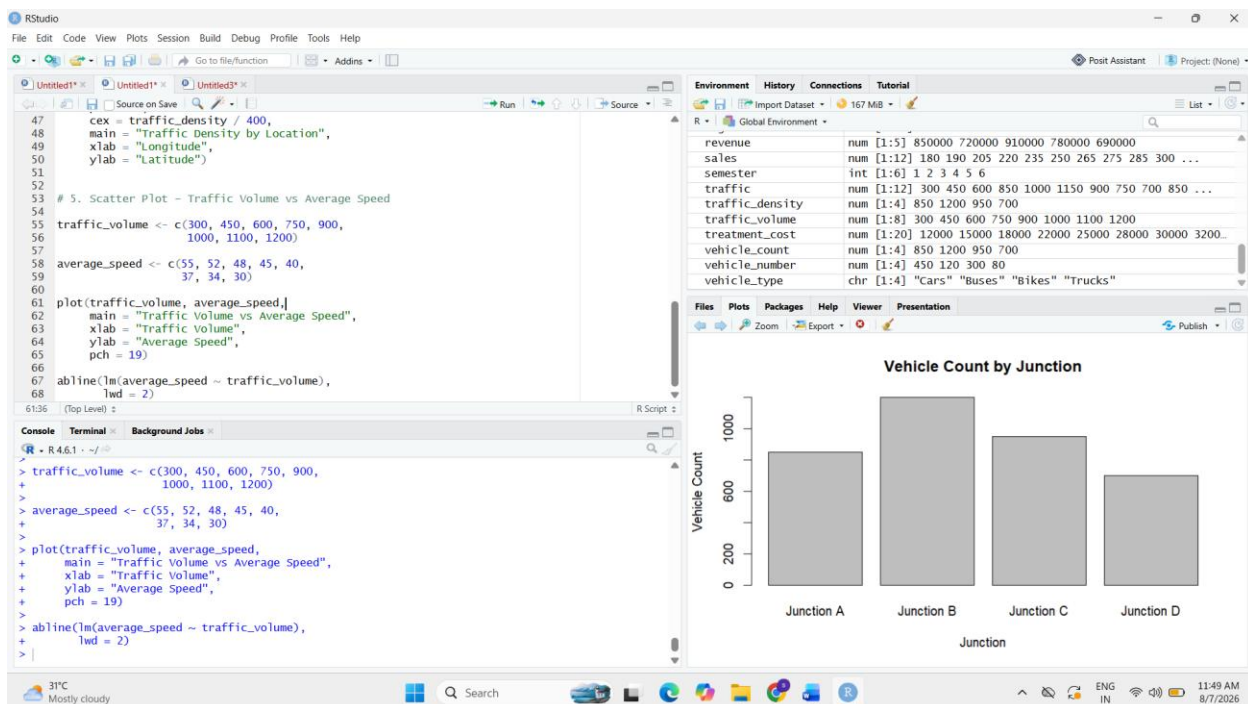
```
junction <- c("Junction A", "Junction B", "Junction C", "Junction D")
```

```
vehicle_count <- c(850, 1200, 950, 700)
```

```
barplot(vehicle_count,
        names.arg = junction,
        main = "Vehicle Count by Junction",
        xlab = "Junction",
        ylab = "Vehicle Count")
```

Explanation:

The bar plot compares the number of vehicles recorded at each junction. Junction B has the highest vehicle count.



2. Line Chart – Hourly Traffic Trend

```
hour <- 6:17
```

```
traffic <- c(300, 450, 600, 850, 1000, 1150,
            900, 750, 700, 850, 1050, 1200)
```

```
plot(hour, traffic,
     type = "o",
     main = "Hourly Traffic Trend",
```

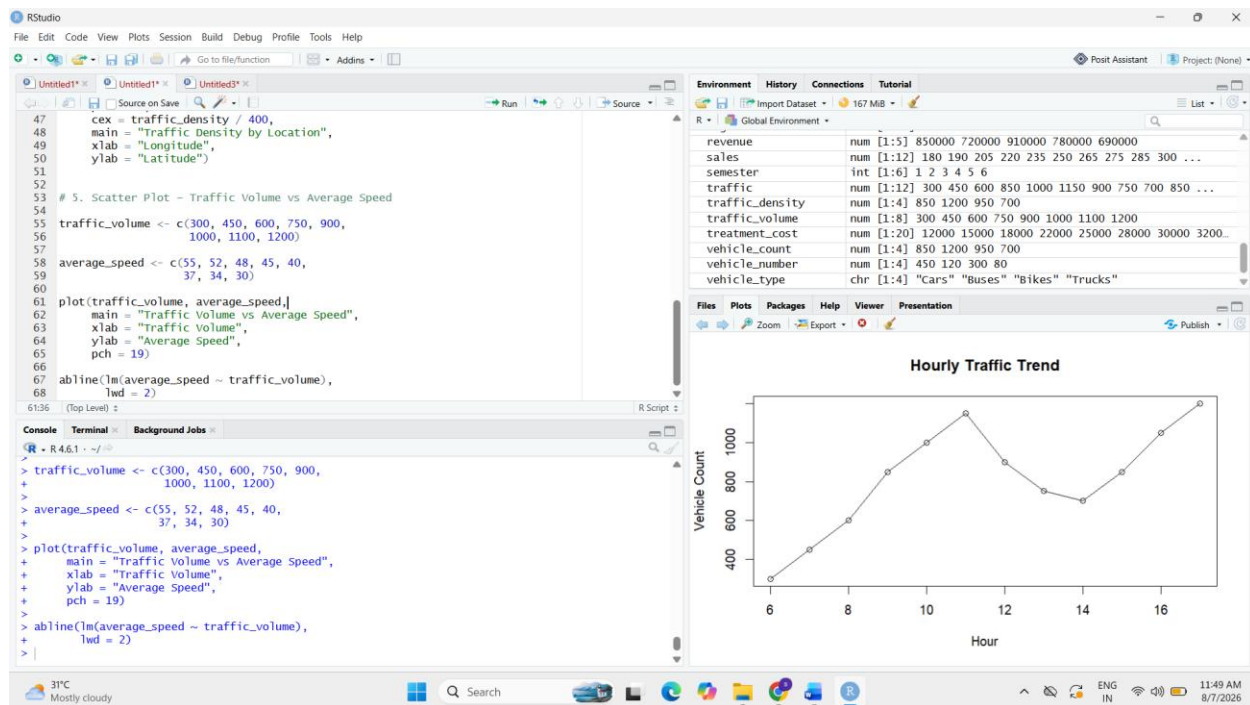
```

xlab = "Hour",
ylab = "Vehicle Count")

```

Explanation:

The line chart shows changes in traffic volume over different hours. Traffic increases during peak periods.



3. Pie Chart – Vehicle Type Proportion

```

vehicle_type <- c("Cars", "Buses", "Bikes", "Trucks")
vehicle_number <- c(450, 120, 300, 80)

```

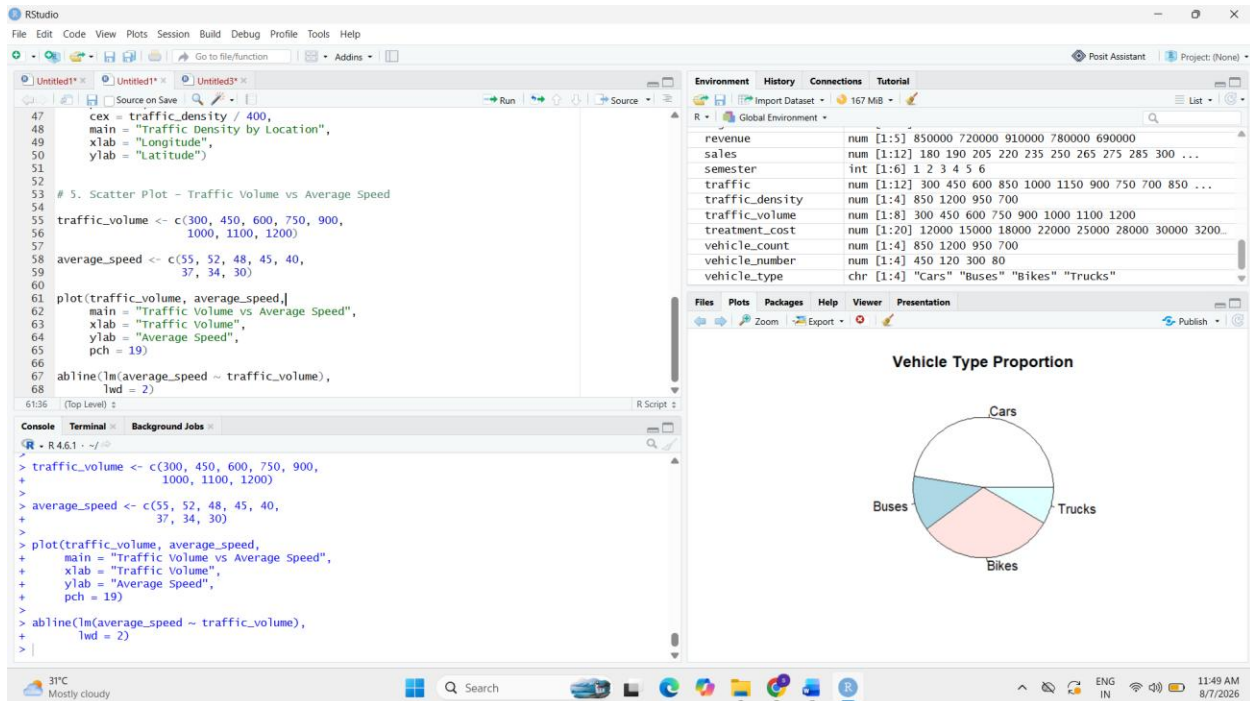
```

pie(vehicle_number,
     labels = vehicle_type,
     main = "Vehicle Type Proportion")

```

Explanation:

The pie chart represents the proportion of different vehicle types. Cars form the largest share of the traffic.



4. Geospatial Map – Traffic Density by Location

```
latitude <- c(13.0827, 13.0878, 13.0750, 13.0900)
```

```
longitude <- c(80.2707, 80.2785, 80.2650, 80.2850)
```

```
traffic_density <- c(850, 1200, 950, 700)
```

```
plot(longitude, latitude,
```

```
  pch = 19,
```

```
  cex = traffic_density / 400,
```

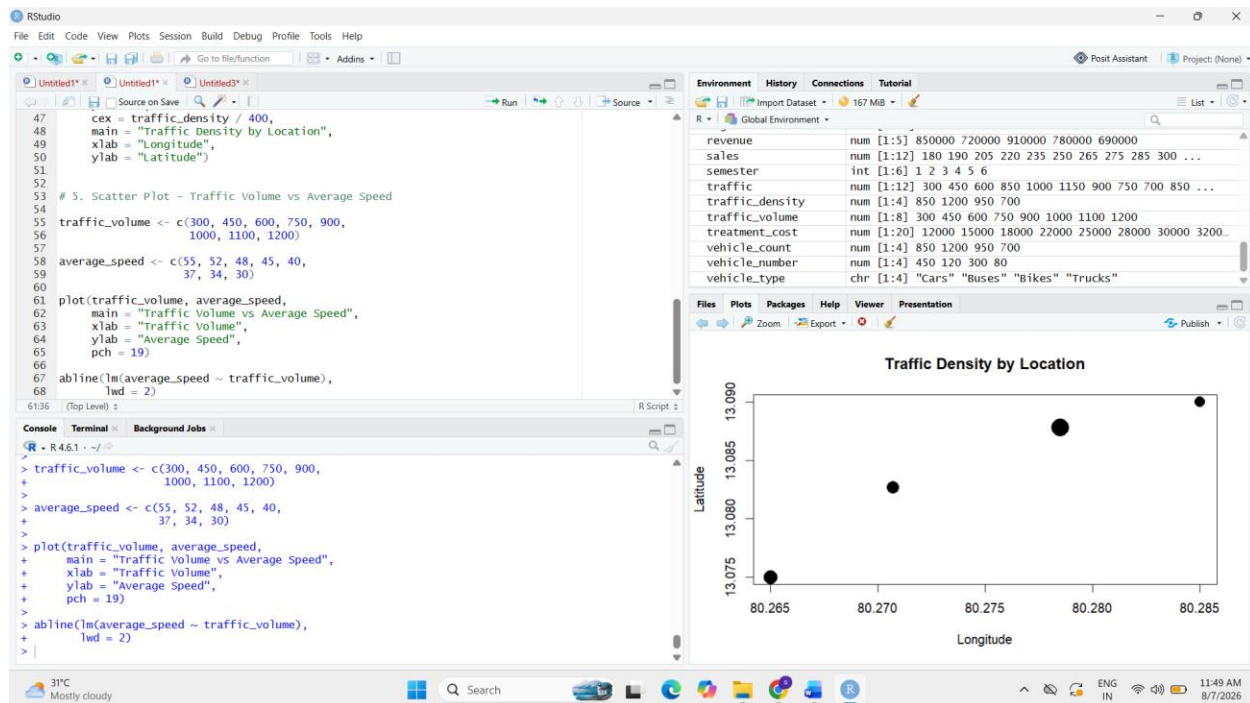
```
  main = "Traffic Density by Location",
```

```
  xlab = "Longitude",
```

```
  ylab = "Latitude")
```

Explanation:

The geospatial plot represents traffic density at different geographical coordinates. Larger points indicate locations with higher traffic density.



5. Scatter Plot – Traffic Volume vs Average Speed

```
traffic_volume <- c(300, 450, 600, 750, 900,
1000, 1100, 1200)
```

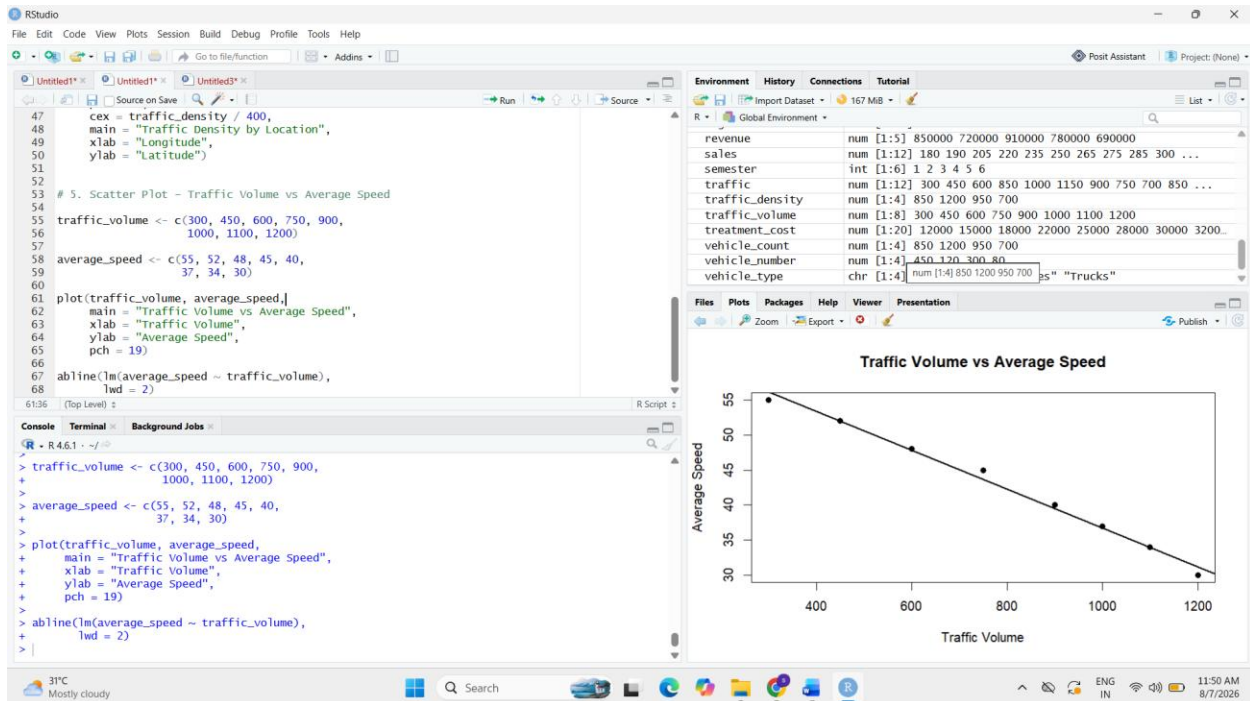
```
average_speed <- c(55, 52, 48, 45, 40,
37, 34, 30)
```

```
plot(traffic_volume, average_speed,
main = "Traffic Volume vs Average Speed",
xlab = "Traffic Volume",
ylab = "Average Speed",
pch = 19)
```

```
abline(lm(average_speed ~ traffic_volume),
lwd = 2)
```

Explanation:

The scatter plot shows that average speed decreases as traffic volume increases.



Task 3: Interpretation

1. Which junction experiences the highest congestion?

Junction B experiences the highest congestion because it has the highest vehicle count of **1,200 vehicles**.

2. During which hours is traffic highest?

Traffic is highest during the **late afternoon/evening peak period**, with the maximum recorded traffic at **5 PM**.

3. Does increased traffic reduce vehicle speed?

Yes. The scatter plot shows a **negative relationship** between traffic volume and average speed. As traffic volume increases, average speed decreases.

Case Study 5 – Social Media User Interaction Analysis

The case study requires visualizations for likes distribution, followers vs engagement, user interaction networks, posting activity over time, and comparison of likes, shares, and comments. It requires a **Histogram with Density Curve, Scatter Plot, Network Visualization, Temporal Line Plot, and Heatmap** in R.

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Justification
Distribution of likes	Histogram with Density Curve	Shows the distribution and concentration of likes.
Followers vs engagement	Scatter Plot	Shows the relationship between followers and engagement.
User interaction network	Network Visualization	Represents connections and interactions between users.
Posting activity over time	Temporal Line Plot	Shows how posting activity changes over time.
Likes, shares and comments comparison	Heatmap	Allows comparison of multiple engagement metrics across users.

Task 2: R Programs

1. Histogram with Density Curve – Distribution of Likes

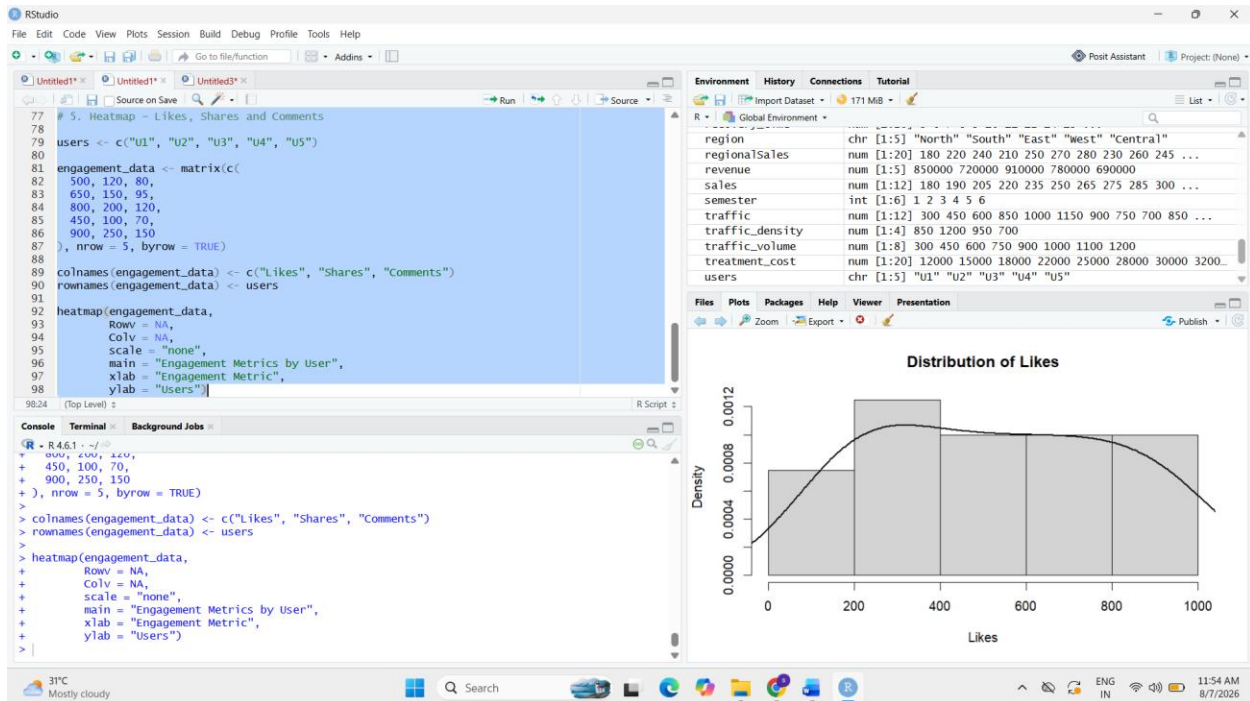
```
likes <- c(120, 150, 180, 220, 250, 300, 350, 400,
          450, 500, 550, 600, 650, 700, 750, 800,
          850, 900, 950, 1000)
```

```
hist(likes,
     probability = TRUE,
     main = "Distribution of Likes",
     xlab = "Likes",
     ylab = "Density")
```

```
lines(density(likes),
      lwd = 2)
```

Explanation:

The histogram shows the distribution of likes, while the density curve provides a smooth representation of the distribution.



2. Scatter Plot – Followers vs Engagement

```
followers <- c(500, 800, 1200, 1600, 2000,
              2500, 3000, 3500, 4000, 5000)
```

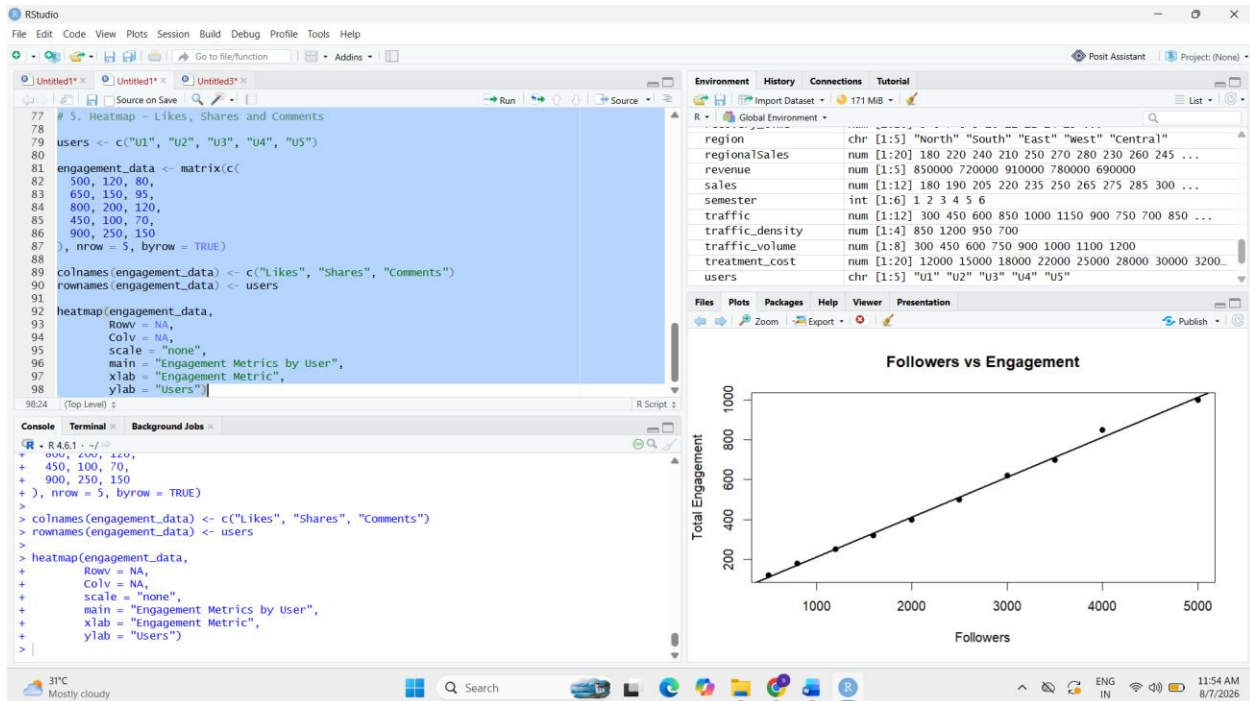
```
engagement <- c(120, 180, 250, 320, 400,
                500, 620, 700, 850, 1000)
```

```
plot(followers, engagement,
     main = "Followers vs Engagement",
     xlab = "Followers",
     ylab = "Total Engagement",
     pch = 19)
```

```
abline(lm(engagement ~ followers),
      lwd = 2)
```

Explanation:

The scatter plot shows the relationship between the number of followers and total engagement. The upward trend indicates that users with more followers generally receive greater engagement.



3. Network Visualization – User Interaction Network

install.packages("igraph") # Run only once if required

library(igraph)

```

connections <- data.frame(
  from = c("U1", "U1", "U2", "U2", "U3", "U3",
           "U4", "U4", "U5"),
  to = c("U2", "U3", "U3", "U4", "U4", "U5",
         "U5", "U1", "U1")
)

```

```
network <- graph_from_data_frame(connections, directed = TRUE)
```

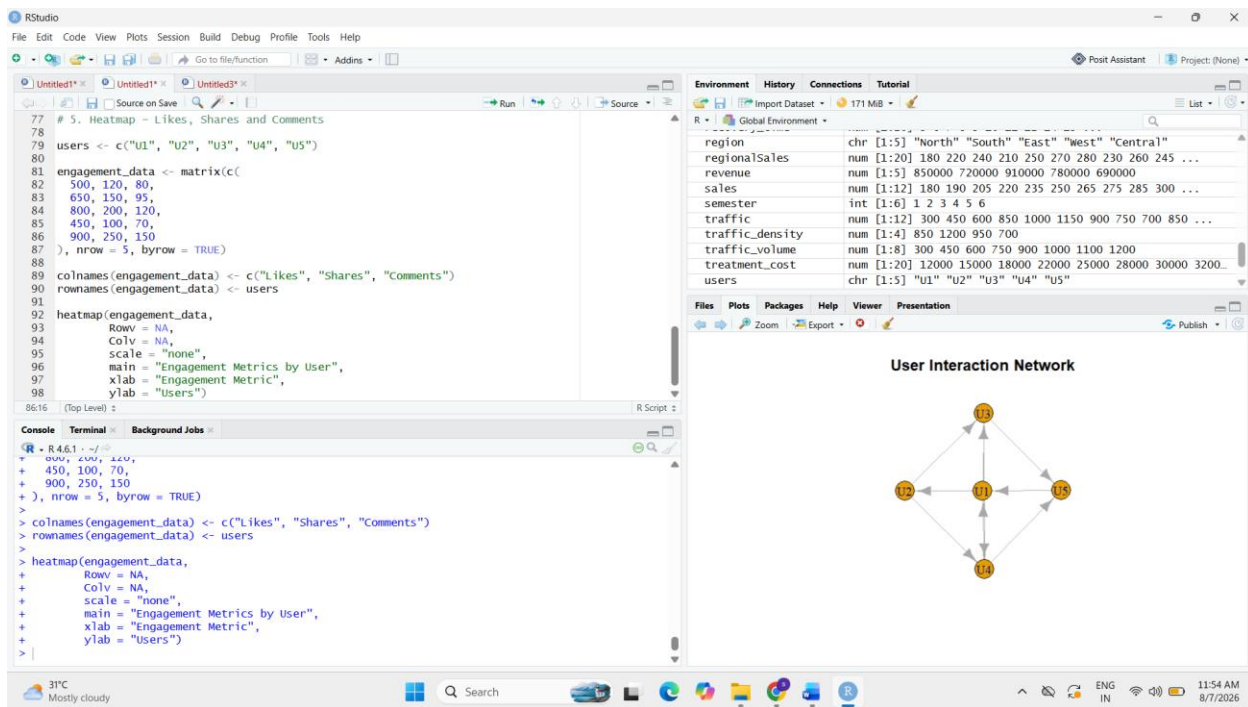
```

plot(network,
  main = "User Interaction Network",
  vertex.size = 25,
  vertex.label.cex = 1)

```

Explanation:

The network visualization represents interactions between users. Users with more connections are more highly connected within the network.



4. Temporal Line Plot – Posting Activity Over Time

```
posting_time <- c("9 AM", "10 AM", "11 AM", "12 PM",  
  "1 PM", "2 PM", "3 PM", "4 PM",  
  "5 PM", "6 PM")
```

```
posts <- c(20, 25, 30, 35, 40, 45, 50, 60, 75, 65)
```

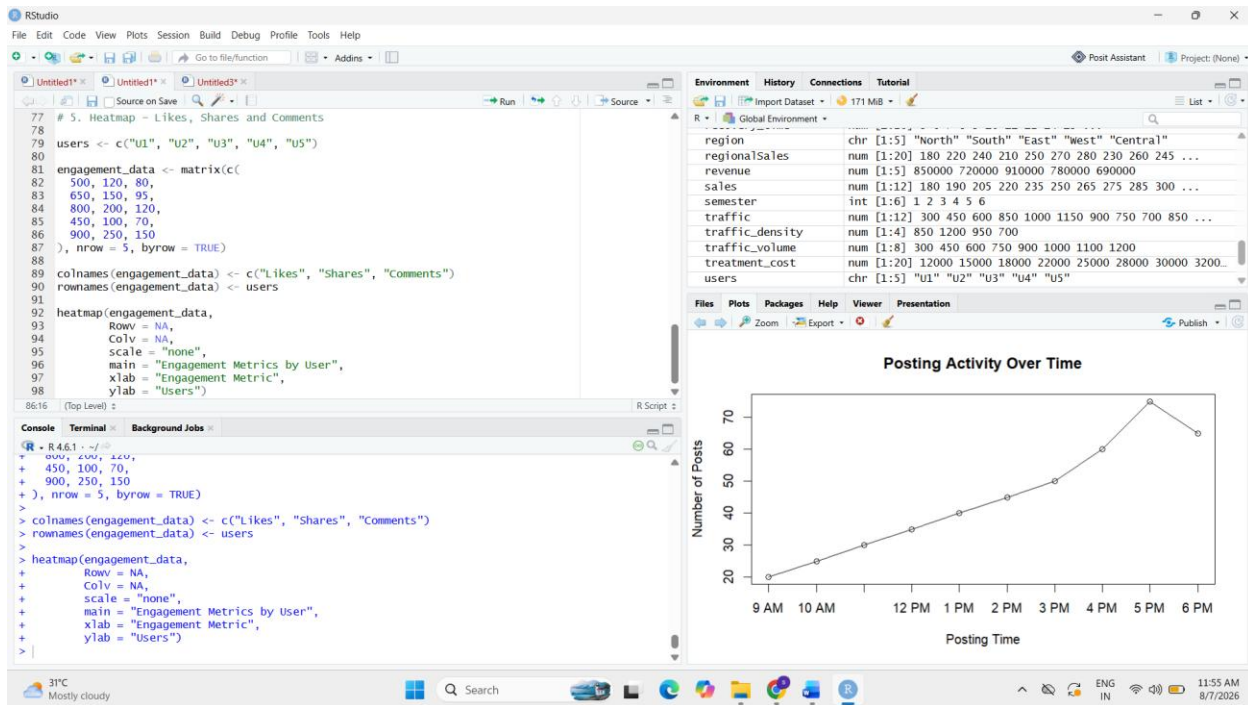
```
plot(1:length(posting_time), posts,  
  type = "o",  
  main = "Posting Activity Over Time",  
  xlab = "Posting Time",  
  ylab = "Number of Posts",  
  xaxt = "n")
```

```
axis(1,
```

```
at = 1:length(posting_time),
labels = posting_time)
```

Explanation:

The temporal line plot shows how posting activity changes during the day. The highest activity occurs around 5 PM.



5. Heatmap – Likes, Shares and Comments

```
users <- c("U1", "U2", "U3", "U4", "U5")
```

```
engagement_data <- matrix(c(
  500, 120, 80,
  650, 150, 95,
  800, 200, 120,
  450, 100, 70,
  900, 250, 150
), nrow = 5, byrow = TRUE)
```

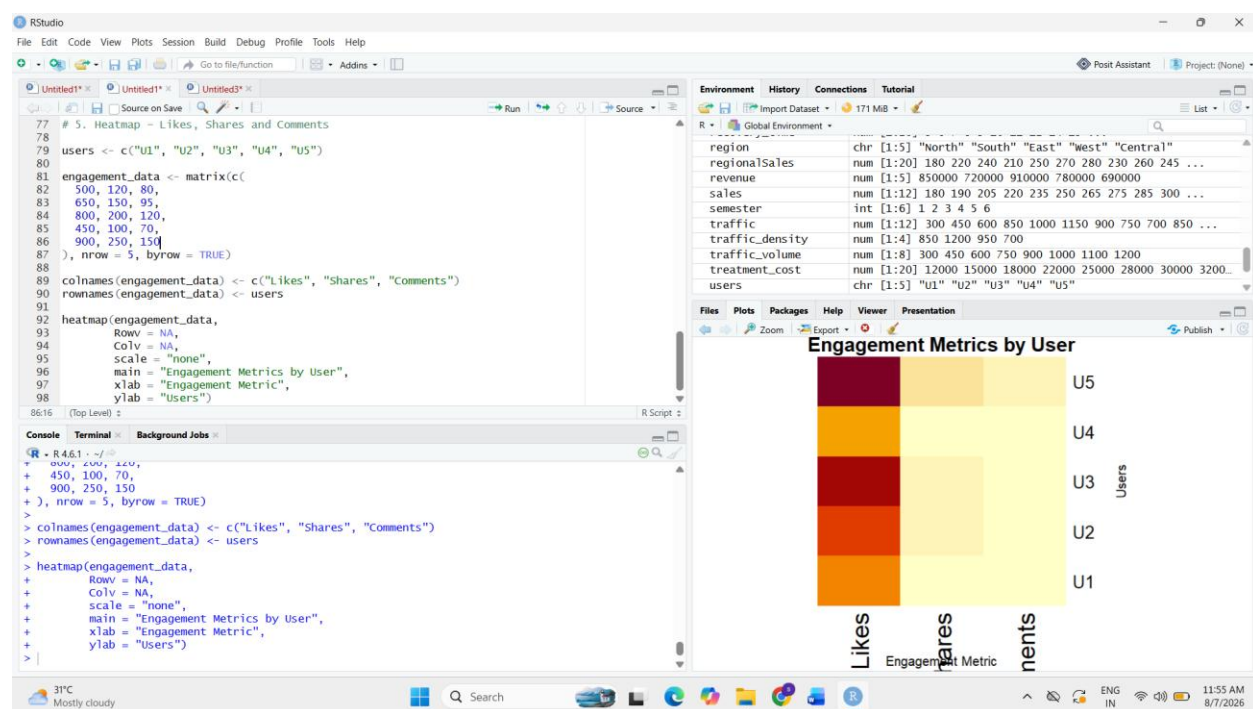
```
colnames(engagement_data) <- c("Likes", "Shares", "Comments")
```

```
rownames(engagement_data) <- users
```

```
heatmap(engagement_data,
        Rowv = NA,
        Colv = NA,
        scale = "none",
        main = "Engagement Metrics by User",
        xlab = "Engagement Metric",
        ylab = "Users")
```

Explanation:

The heatmap compares likes, shares, and comments for each user. It helps identify users with higher engagement levels.



Task 3: Interpretation

1. Which users are the most influential?

U5 is the most influential user because it has the highest engagement values and strong connections within the network.

2. At what time is engagement highest?

Engagement is highest around 5 PM, where posting activity reaches its maximum.

3. Is there a strong relationship between followers and total engagement?

Yes. The scatter plot shows a **strong positive relationship**. As the number of followers increases, total engagement also tends to increase.

4. Identify highly connected users from the network visualization.

U1 and **U3** are highly connected users based on the interaction network.

5. Which engagement metric varies the most across users?

Likes show the greatest variation across users compared with shares and comments.

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